LAKSHMI KANTH PULIPAKA

Bhimavaram, India, +91 9505458711, kanthpulipaka@gmail.com, GitHub, LinkedIn

PROFESSIONAL SUMMARY

Pre-final-year **AI and ML** student at Vishnu Institute of Technology with an **8.78 CGPA**, specializing in **Software Development**, **AI Engineering**, and **Data Science**. Built **Machine Learning**, **Deep Learning**, and **Computer Vision** solutions through research internships and **hackathons**.

SKILLS

Software Development: Python, Java, SQL, DSA, Git, GitHub, Object-Oriented Programming, FastAPI, REST APIs, Scalable System Design, Streamlit.
AI Engineering and Data Science: Supervised and Unsupervised Learning, Classification, Regression, Random Forests, Gradient Boosting, SVM, Hyperparameter Tuning, CNNs, TF-IDF, NLP, Scikit-learn, TensorFlow, End-to-End Model Deployment.
Math and Data Analytics: Probability Theory, Statistical Inference, Bayesian Statistics, Linear Algebra, Hypothesis Testing, EDA, Feature Engineering, Dimensionality Reduction, Pandas, NumPy, Power BI.
Computer Vision: OpenCV, YOLOv8, MediaPipe, Image Processing, Real-Time Detection Systems.
Tools: Jupyter Notebook, VS Code, Git/GitHub, Agile Collaboration.
Core Strengths: Problem Solving, Product-Oriented Thinking, Innovation, Communication, Teamwork, Time Management, Rapid Prototyping.

WORK EXPERIENCE

Summer Research Intern – VLED Lab, IIT Ropar *May 2026 – Present*
- Conducted research on **supervised learning** pipelines, benchmarking **classification** models using **ROC-AUC**, **F1-score**, and **MSE** to identify optimal estimators for real-world datasets.
- Collaborating with a team of researchers and mentors to develop and contribute to **open-source ML software**, accelerating research delivery on **graph-based ML** for large-scale relational data.

Data Science Intern – Developers Arena *Dec 2025 – Mar 2026*
- Improved **model accuracy by 15%** by building **classification** and **regression** pipelines using **Random Forests**, **Gradient Boosting**, and **SVM** with systematic **hyperparameter tuning**.
- Reduced data preparation time by **40%** through automated, **end-to-end** cleaning workflows; applied **EDA** and **feature engineering** across 50K+ record datasets to drive **data-driven decision making**.

PROJECTS

Delivery ETA Optimization using Graph-Network Intelligence *Python, Graph Analytics, Streamlit*
- Built a **scalable**, real-world logistics platform using **graph analytics** and **probabilistic forecasting** to predict delivery ETAs and detect bottleneck hubs, achieving over **96% accuracy** using a **Gradient Boosting** algorithm.
- Implemented **Dijkstra/A*** route optimization with SLA breach monitoring and congestion monitoring in a **Streamlit** operational dashboard. [GitHub]

BedForecast: Hospital Bed Forecasting Platform *scikit-learn, TensorFlow, Streamlit*
- Built three **regression** models – **XGBoost** (ICU), **Random Forest** (general ward), and **Prophet** (emergency) – for **time-series** bed demand forecasting, achieving **90% average accuracy**.
- Developed a live occupancy dashboard with **surge prediction** and automated demand alerts to support proactive bed allocation, validated using **RMSE** and **MAE**. [GitHub]

Campus Eye: AI Surveillance System *YOLOv8, OpenCV, FastAPI*
- Built a real-time traffic violation **detection** and **classification** system using **YOLOv8** with confidence-threshold filtering for scalable, production-grade deployment.
- Automated violation logging with event snapshots, reducing manual review overhead through a structured **rule-based decision pipeline** and **practical, product-focused engineering**. [GitHub]

ACHIEVEMENTS

Google BigCode Program: Selected for demonstrated **DSA**, **algorithmic problem-solving**, and coding proficiency.
ICAT: Qualified with **All India Rank 2139** assessed on mathematical aptitude, logical reasoning, and programming.
Aspire Leaders Program: Finalist in global initiative on AI ethics and responsible ML deployment.

CERTIFICATIONS

AWS AI and ML Scholars Challenge (AWS and Udacity): Machine Learning Engineering, Deep Learning, and applied AI project development.
Google Advanced Data Analytics Professional Certificate: Statistics, Hypothesis Testing, Regression, Probability.
IBM – Machine Learning with Python: Supervised Learning, Classification, Clustering, Regression.
IBM SkillsBuild – Getting Started with Data.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning Expected 2028
Vishnu Institute of Technology, Bhimavaram CGPA: 8.78/10
Intermediate (Class XII) 2024
Sri Sarada Junior College 96.8%.